INCH-POUND

MIL-DTL-16878/16C
11 August 2000
SUPERSEDING
MIL-W-16878/16B(NAVY)
11 September 1992

DETAIL SPECIFICATION SHEET

WIRE, ELECTRICAL, CROSSLINKED, MODIFIED POLYETHYLENE (XLPE) INSULATED, 125 °C, 3000 VOLTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16878G.

REQUIREMENTS.

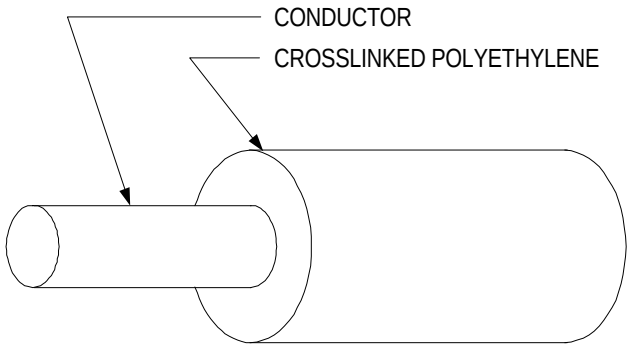

FIGURE 1. Wire configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Wire configuration and dimensions.

PIN 1/	Wire Size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material	Coating		Min	Max
M16878/16BDA*	26	1 X 26	Copper	Tin	.0159	.067	.082
M16878/16BDB*	26	7 X 34	Copper	Tin	.0190	.070	.085
M16878/16BDE*	26	19 X 38	Copper	Tin	.0200	.070	.085
M16878/16BEA*	24	1 X 24	Copper	Tin	.0201	.071	.086
M16878/16BEB*	24	7 X 32	Copper	Tin	.0240	.075	.090
M16878/16BEE*	24	19 X 36	Copper	Tin	.0250	.075	.090
M16878/16BFA*	22	1 X 22	Copper	Tin	.0254	.076	.091
M16878/16BFB*	22	7 X 30	Copper	Tin	.0300	.081	.098
M16878/16BFE*	22	19 X 34	Copper	Tin	.0320	.081	.098
M16878/16BGA*	20	1 X 20	Copper	Tin	.0320	.083	.098
M16878/16BGB*	20	7 X 26	Copper	Tin	.0380	.089	.105
M16878/16BGE*	20	19 X 32	Copper	Tin	.0400	.089	.105
M16878/16BHA*	18	1 X 18	Copper	Tin	.0403	.091	.106
M16878/16BHB*	18	7 X 28	Copper	Tin	.0480	.099	.114
M16878/16BHE*	18	19 X 30	Copper	Tin	.0470	.099	.114
M16878/16BJA*	16	1 X 16	Copper	Tin	.0508	.102	.117
M16878/16BJE*	16	19 X 29	Copper	Tin	.0570	.108	.123
M16878/16BJF*	16	26 X 30	Copper	Tin	.0600	.111	.125
M16878/16BKA*	14	1 X 14	Copper	Tin	.0641	.115	.130
M16878/16BKE*	14	19 X 27	Copper	Tin	.0720	.123	.138
M16878/16BKH*	14	41 X 30	Copper	Tin	.0760	.127	.142
M16878/16BLE*	12	19 X 25	Copper	Tin	.0920	.156	.171
M16878/16BLG*	12	37 X 28	Copper	Tin	.0890	.153	.168
M16878/16BLJ*	12	65 X 30	Copper	Tin	.0930	.157	.172
M16878/16BMG	10	37 X 26	Copper	Tin	.1110	.174	.189
M16878/16BNL	8	133 X 29	Copper	Tin	.1670	.239	.254
M16878/16BNM	8	168 X 30	Copper	Tin	.1690	.239	.254
M16878/16BPL	6	133 X 27	Copper	Tin	.2100	.283	.298
M16878/16BRL	4	133 X 25	Copper	Tin	.2680	.346	.366
M16878/16BSL	2	133 X 23	Copper	Tin	.3550	.415	.435
M16878/16BSP	2	665 X 30	Copper	Tin	.3420	.415	.435
M16878/16BTN	1	259 X 25	Copper	Tin	.3780	.468	.488
M16878/16BTR	1	817 X 30	Copper	Tin	.3820	.468	.488
M16878/16BUN	0	259 X 24	Copper	Tin	.4240	.518	.538
M16878/16BUS	0	1045 X 30	Copper	Tin	.4310	.518	.538
M16878/16BWT	00	1330 X 30	Copper	Tin	.4750	.575	.599
M16878/16BYV	000	1672 X 30	Copper	Tin	.5350	.657	.681
M16878/16BZW	0000	2109 X 30	Copper	Tin	.6000	.790	.733

Notes:
1/ PIN stands for part or identifying number (see figure 2).

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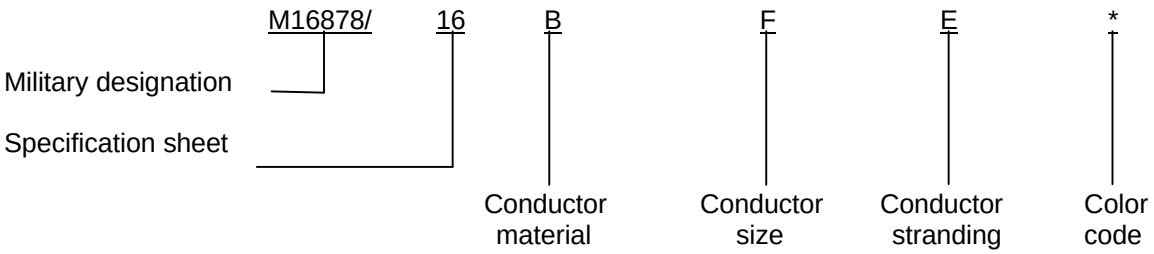

FIGURE 2. Example of PIN (see MIL-DTL-16878G).

Configuration and dimensions:	See figure 1 and table I
Operating voltage:	Up to 3000 volts
Operating temperature:	Up to 125 °C
Insulation:	Crosslinked polyethylene
Spark test voltage:	8.0 kV
Impulse dielectric test voltage:	12.0 kV, or 8.5 kV using the 3.0 kHz spark test
Dielectric withstanding voltage:	6.0 kV
Partial discharge voltage:	Required
Insulation resistance:	$IR = K \log_{10} D/d$
	Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20 °C
	K = 10,000
	D = Maximum average diameter of finished wire
	d = Conductor diameter
Cold bend:	Condition 4 hours at -55±1 °C (see table II)

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel diameter (inches, maximum)
26 through 22	2
20 through 14	3
12, 10	4.5
8	6
6, 4	10
2, 1, 0	18
00, 000, 0000	20

Surface resistance:	Not required
Heat resistance:	Condition at 150 °C
Heat aging:	25 percent change (maximum) in 96 hours at 135 °C
Insulation tensile strength:	1800 pounds force per square inch (minimum)
Insulation elongation:	100 percent (minimum)
Partial discharge test:	Required
Marking and stripe durability:	Not required

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy - SH
Air Force - 11
DLA - CC

Review activity:
Navy - AS

Preparing activity:
DLA - CC

(Project 6145-2193-009)

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